

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fifth Semester of B. Tech (IT) Examination****March 2018****IT316 ADVANCED JAVA PROGRAMMING (ELECTIVE-I)****Date: 29.03.2018, Thursday****Time: 01:30 p.m. To 04.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Do as directed: [07]**

- a. Explain every phase of JSP life cycle. (03)
- b. Explain request and response implicit objects with example. (04)

Q – 2.a Explain doGet() and doPost() methods available in servlet. [04]**Q – 2.b Explain following JSP action elements or action tags with example. [04]**

1. `<jsp:include>`
2. `<jsp:forward>`

OR**Q – 2.b Explain following JSP directives with example. [04]**

1. `<%@ page ... %>`
2. `<%@ include ... %>`

Q – 2.c Explain different types of JDBC drivers. [06]**OR****Q – 2.c Explain servlet life cycle with Architecture Diagram. [06]****Q - 3 Answer any TWO. [14]**

- a. Write a program to make counter application in servlet. When the counter value reaches to 100, servlet page will be redirected to another servlet which display the welcome message for the user.
- b. Write a program in JSP to demonstrate the use of `<jsp:usebean>` action element.
- c. Write a Servlet that take First_ Name & Last_ Name from the user & generate login-id such that two characters are from First_ Name & four characters are from Last_ Name. Display the generated login-id.

SECTION – II**Q - 4 Do as directed: [07]**

- a. Write the steps for executing RMI program. (04)
- b. Explain ServerSocket with example. (03)

Q – 5.a Write a java program to send the mail to the Gmail user with file attachment using mail API. [03]**Q – 5.b Explain following JSTL core library tags. [06]**

1. `<c:out>`
2. `<c:if>`
3. `<c:set >`

OR**Q – 5.b Explain following JSTL sql library tags. [06]**

1. `<sql:setDataSource>`
2. `<sql:query>`
3. `<sql:param>`

Q – 5.c Explain bind() and accept() methods of ServerSocket class. [05]**OR**

Q – 5.c Explain stub and skeleton in RMI with diagram and example. **[05]**

Q – 6 **Attempt any Two:** **[14]**

- a. Write Swing application to convert temperature from Celsius to Fahrenheit.
- b. Write a program to make a calculator in RMI which perform the basic operations like addition, subtraction, multiplication and division.
- c. Write a socket program to send “Hello!” message to client machine using port 5555.
